

Question number	Answer	Additional guidance	Mark
6(a)(i)	Choose an item. • C is the correct answer A is not the correct answer as adrenaline does not decrease the breathing rate. B is not the correct answer as adrenaline does not decrease the heart rate. D is not the correct answer as adrenaline does not decrease the breathing rate or heart rate.		(1)

Question number	Answer	Additional guidance	Mark
6(a)(ii)	An answer that includes three of the following points: • (mean) heart rate decreases as nicotine concentration increases (with or without adrenaline) (1) • (mean) heart rate increases as adrenaline concentration increases (1) • highest (mean) heart rate is with {highest / 0.1mmol dm^3} adrenaline concentration and {lowest / 0.001mmol dm^3} nicotine concentration (1) • lowest (mean) heart rate is with {0.001mmol dm^3} adrenaline concentration and {highest / 1000mmol dm^3} nicotine concentration (1) • highest adrenaline concentration and no nicotine caused death of Daphnia (1)	Accept converse accept nicotine is a depressant accept adrenaline is a stimulant accept {0.00 / control / nil} for lowest nicotine concentration	(3)

Question number	Answer	Additional guidance	Mark
6(b)	Choose an item. • C is the correct answer A is not the correct answer as a gibberellin is not an enzyme that converts starch to glucose B is not the correct answer as a gibberellin is not a molecule that converts Pfr to Pr D is not the correct answer as a gibberellin is not a protein that wraps around DNA		(1)

Question Number	Answer
*6(c) 1 2 3	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material indicated as relevant. Additional content included in the response must be scientific and relevant. accept R for red light and FR for far red light FROM SECTIONS 1 TO 3 MAX 1 credit FROM EACH comment on short day (long night) plants (from the diagram) • short day plants generally require {light period of less than 12 hours / dark period 8-10 hours / longer than critical night length} for flowering • description of light flashes and subsequent flowering or non flowering • interruption of critical night length by {1×R / 2×R and 1×FR} light periods prevents flowering • interruption of critical night length by {1×R and 1×FR / 2×R and 2×FR} light periods gives flowering • there is no flowering if last light flash is R during dark period / the plant flowers if last flash of light is FR • comment on long day (short night) plants (from the diagram) • long day plants require light periods {of 14-16 hours / longer light periods / shorter dark periods / shorter than critical night length} for flowering • critical night length (with no light flashes) prevents flowering • (long day) plants flower {with any (brief) R in long dark night / if last flash of light is R} / interruption of critical night length by {one R (660nm) / two R and one FR} light periods allows flowering • interruption of critical night length by {1×R + 1×FR / 2×R and 2×FR} light periods prevents flowering • comment on gibberellin graph • number of flowers affected by gibberellin concentration / the number of flowers increase and then decreases as gibberellin increases • {intermediate concentration / 1.5 µm} gave greatest number of invitro flowers • highest concentration / 4.4 µm} gave least effect • description of pattern of numbers of flowers with GA concentration • comment on error bars and validity

FROM THESE SECTIONS BELOW MAX 2 credit FROM EACH

4+ 5

- **comment on gibberellins mode of action**

gibberellins may promote or inhibit flowering

GA affects metabolic and regulatory pathways

GA action on aleurone layer and secretion of amylase (accept in relation to seed as well)

GA acting as a transcription factor

GA affect gene expression / cell division linked to flowering

- **comment on phytochromes**

{photoreceptors / phytochromes} absorb (red and far red) light

{Pfr low / Pr high} gives flowering in short day plants

{Pfr high / Pr low} gives flowering in long day plants

active form Pfr can activate / repress specific gene expression linked to flowering

Pfr-----far red light (730nm)---→ Pr (inactive)

Pr-----red light (660nm)-----→ Pfr (active form)

Full sunlight contains more red than far red light so at sundown all phytochrome is Pfr

{in light / during the day} Pr is converted into Pfr

In the dark Pfr spontaneously converts back to Pr

reference to photoperiodism – importance of relative duration of {light / dark} periods

6 + 7

(6)

		Additional guidance	
Level 0	0	No awardable content	
Level 1	1-2	Level 1: Description of (a minimum of 2 specific) results from diagram and graph	
Level 2	3-4	Level 2: Reference to all of level one plus limited discussion of the effects of light on flowering and /or gibberellin	
Level 3	5-6	Level 3: Reference to all of level one and two one detailed discussion of the effects of light on flowering and gibberellin	

